A local obstruction to quadratic input maps

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Abstract

Consider the five-contact polynomial $r(T) = T(T-1)(T-2401)(32T+343)(2T-1)$. No integer polynomial q of degree at most two can simultaneously reflect integrality over the rationals, take odd values at every integer, and make $r(q(x))$ a square in $\mathbb{Q}_2$ for every integer x. The obstruction follows from an exact classification of quadratic integrality-reflecting maps and elementary two-adic calculations. It concerns this particular scalar normalization and imposes no lower bound on the degree of an arbitrary Diophantine definition.

1 Integrality-reflecting quadratic maps

Say that $q \in \mathbb{Z}[X]$ reflects rational integrality if

$$q(x) \in \mathbb{Z} \quad \Longleftrightarrow \quad x \in \mathbb{Z} \qquad (x \in \mathbb{Q}).$$

Proposition 1.1. *Let $q(X) = AX^2 + BX + C$ with nonzero integer A. Then q reflects rational integrality if and only if $|A|$ is squarefree and A divides B.*

Proof. Since $q(-B/A) = C$, reflection forces $A \mid B$. Write $B = Ak$. If a prime square p^2 divides A, then

$$q(1/p) = A/p^2 + Ak/p + C \in \mathbb{Z},$$

contrary to reflection. Thus $|A|$ is squarefree.

Conversely, suppose these two conditions hold. Write $x = u/v$ in lowest terms, with $v > 0$. If $q(x)$ is integral, then

$$v^2 \mid Au(u + kv).$$

Both u and $u + kv$ are prime to v, so $v^2 \mid A$. Squarefreeness gives $v = 1$. □

2 Three nonsquare residue classes

Put

$$r(T) = T(T-1)(T-2401)(32T+343)(2T-1).$$

For a nonzero rational number a, write $v_2(a)$ for its normalized two-adic valuation. An odd integer is a square in $\mathbb{Q}_2$ precisely when it is congruent to one modulo eight; in general the valuation must also be even.

Lemma 2.1. *For every integer t in one of the following classes, $r(t)$ is a nonzero nonsquare in $\mathbb{Q}_2$:*

Class of t	$v_2(r(t))$	$r(t)/2^{v_2(r(t))} \pmod 8$
$t \equiv 5 \pmod 8$	4	3
$t \equiv 7 \pmod 8$	2	5
$t \equiv 9 \pmod{16}$	6	3

Proof. The two even factors differ by

$$(t-1)-(t-2401)=2400=2^5\cdot 75.$$

If $t\equiv 5 \pmod 8$, both have valuation two; their odd parts differ by $600\equiv 0 \pmod 8$, so their product is one modulo eight. The other three factors have product $5\cdot 7\cdot 1\equiv 3 \pmod 8$. If $t\equiv 7 \pmod 8$, both even factors have valuation one; their odd parts differ by $1200\equiv 0 \pmod 8$. The other factors have product $7\cdot 7\cdot 5\equiv 5 \pmod 8$.

If $t\equiv 9 \pmod{16}$, both even factors have valuation three. Their odd parts differ by $300\equiv 4 \pmod 8$; the product of those odd parts is therefore five modulo eight. The other three factors have product $1\cdot 7\cdot 1\equiv 7 \pmod 8$, and $5\cdot 7\equiv 3 \pmod 8$. □

3 The obstruction

Theorem 3.1. *There is no polynomial $q\in\mathbb{Z}[X]$ of degree at most two satisfying all three conditions:*

1. *q reflects rational integrality;*
2. *$q(x)$ is odd for every $x\in\mathbb{Z}$;*
3. *$r(q(x))$ is a square in $\mathbb{Q}_2$ for every $x\in\mathbb{Z}$.*

Here zero may be included among the squares; this convention does not affect the result.

Proof. A constant polynomial cannot reflect integrality. A linear polynomial $BX+C$ reflects integrality only when $B=\pm 1$: otherwise its value at $1/B$ is integral with nonintegral input. For $B=\pm 1$ its integer values have both parities. Thus assume $q(X)=AX^2+BX+C$ with $A\neq 0$.

By Proposition 1.1, A is squarefree up to sign and $B=Ak$. Odd integer values imply C odd and $A(1+k)$ even.

If A is odd, then k is odd. The values of $x(x+k)$ modulo eight are all four even residues. Indeed, replacing x by $x+4$ changes its value by four modulo eight, and replacing x by $x+2$ changes it by two modulo four. Multiplication by the odd number A therefore makes $q(x)$ attain every odd residue modulo eight. In particular it attains seven, contrary to Lemma 2.1.

We may therefore write $A=2a$ with a odd. If k is odd, $x(x+k)$ attains both zero and two modulo four. Thus $q(x)$ attains both C and $C+4$ modulo eight. One of these is five or seven, again a contradiction.

It remains to consider $k=2h$. Completing the square gives

$$q(x)=A(x+h)^2+c,\qquad c=C-Ah^2 \text{ odd}.$$

Translation by the integer h preserves the set of integer inputs, so it is enough to consider $q(y)=Ay^2+c$. If c is five or seven modulo eight, take $y=0$. Otherwise c is one or three. Testing odd y shows that $c+A$ must also be one or three modulo eight. Since $A\equiv 2$ or $6 \pmod 8$, the only remaining possibilities are

$$(c,A)\equiv(1,2) \pmod 8 \quad\text{or}\quad (c,A)\equiv(3,6) \pmod 8.$$

In the first case, the values at $y=0$ and $y=2$ differ by eight modulo sixteen; one is nine modulo sixteen and is excluded by Lemma 2.1.

In the second case $c+A$ is one or nine modulo sixteen. The latter is excluded by $y=1$. In the former case,

$$\frac{1-c}{A}\in\mathbb{Z}_2^\times,\qquad \frac{1-c}{A}\equiv 1 \pmod 8.$$

There is therefore an odd $y_0 \in \mathbb{Z}_2$ with $Ay_0^2 + c = 1$. Choose an integer y with $y - y_0 \equiv 16 \pmod{32}$. Then $v_2(y - y_0) = 4$ and $v_2(y + y_0) = 1$, giving

$$v_2(q(y) - 1) = v_2(A) + 4 + 1 = 6.$$

Since $v_2(2400) = 5$, it follows that $v_2(q(y) - 2401) = 5$. The other three factors of $r(q(y))$ are odd, so

$$v_2(r(q(y))) = 11.$$

This odd valuation is incompatible with being a two-adic square. Every possible case has been excluded. □

Corollary 3.2. *Among integer polynomials satisfying the three conditions of Theorem 3.1 and $q(n) > 0$ for every $n \in \mathbb{Z}$, the minimum degree is four. It is attained by $q(X) = 8X^4 + 3$. If $q(n) > 2401$ is required, $q(X) = 8X^4 + 2403$ attains the same degree.*

Proof. A nonconstant polynomial positive on all integers has even degree. The theorem excludes degree two, and constants do not reflect integrality. For either displayed quartic, integrality at a reduced rational input u/v implies $v^4 \mid 8$, and hence $v = 1$. Its values have the required positivity and are three modulo eight. For every integer $t \equiv 3 \pmod 8$, the two even factors of $r(t)$ have valuation one and odd parts differing by $1200 \equiv 0 \pmod 8$. Their odd-part product is one modulo eight, as is the product $3 \cdot 7 \cdot 5$ of the other three factors. Thus $v_2(r(t)) = 2$ and $r(t)/4 \equiv 1 \pmod 8$, proving that $r(t)$ is a square in $\mathbb{Q}_2$. □

The conclusion concerns the simultaneous requirements in the theorem. It does not exclude quadratic input maps when parity is unrestricted, when an additional scalar correction is used, or when a different contact polynomial or pole construction is chosen. It therefore identifies a restriction of the stated normalization, rather than a minimum degree for the overall research problem.

4 Exact finite checks

The standard-library program `check_quadratic_input_obstruction.py` checks all residue representatives modulo 512 in Lemma 2.1. It also checks the four images of $x(x + k)$ for odd k modulo eight. For the final case of the proof it examines every pair

$$A \equiv 6 \pmod 8, \qquad c \equiv 3 \pmod 8, \qquad A + c \equiv 1 \pmod{16}$$

modulo 128, and explicitly finds $0 \le y < 32$ with

$$Ay^2 + c - 1 \equiv 64 \pmod{128}.$$

These 128 cases all have the asserted valuations six, five, and eleven. The accompanying JSON certificate lists each witness. The checker also verifies the square residue class $t \equiv 3 \pmod 8$ and the two quartic examples in Corollary 3.2. The proof is elementary and does not depend on any arithmetic hypothesis from a defining-polynomial construction.